GITHUB LINK: https://github.com/viio00/NEU-LMS
Create an NEU Library Visitor Log app

KEY POINTS

1. Visitors: Student, faculty (employees), guests(outsiders)
2. Students and employees are stored in database (dummy accounts) with their information
3. Guests will either have a QR code (pre-registration) or fill up a form
4. Admins (role-based switching): jcesperanze@neu.edu.ph, cardonachynnam@gmail.com, chynna.cardona@neu.edu.ph
5. Reports can be downloaded into pdf file
6. Statistics/reports can be filtered by day (day, week, month, year), department/program, purpose, and who (student/faculty)

(For testing, go to last page)
(See full UI functions)

Initial prompt

I. Into Page
- Enter user input of ID No. Format: XX-XXXXX-XXX
- OR scan ID/ QR for outsiders (for QR testing, well be using a drop image and then scan it internally to get info)
- Should contain welcome message
- Time and date (live)
- Then upper dashboard for logo, title: New Era University Library
- Visitor mode (which is this page) and Admin page button

III. After that, it will redirect to "purpose page"
- Research/thesis
- Study
- Use of Computer
- Boardgames/ pass time
- Others

III.Visitor GUI
- Name
- Program Course (if student), Faculty department (if faculty)
- date&time (in/out)
- stud no./emp no.
- Purpose

FOR OUTSIDERS: Name, From what university, occupation or student, address, contact no. (info based on qr filled up)

IV: Admin dashboard GUI

- Institutional email and password (button for log in will light up/ 100% opacity when pass and email is inputted)
- statistics on: Visitors per day, week, month, ave. duration (stay), “purpose” stats
- get hours (hours_visited = out-in)
- activity log (info of visitors)
- In activity log, admins can tap profile and see info (in visitors)

Users: admin/visitor
Admin - can block visitor through app

Output: if success, else error (back to homepage)
1. Name, dep't(output) - purpose(pre-selected)
2. Message intro
3. FOR ADMIN: Dashboard
4. Reports can be generated/download into a pdf file

All passwords in admin will be set to = “passW@rd”
Adimns: admin1@neu.edu.ph, admin2@neu.edu.ph, admin3@neu.edu.ph

Key Features Implemented:

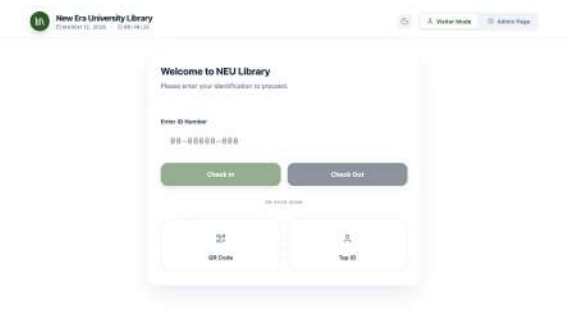
1. **Visitor Entry System**:
    - **ID Input**: Supports the standard XX-XXXXX-XXX format for students and faculty.
    - **QR Scan Simulation**: Allows users to upload an image containing a QR code (simulated via jsqr) to automatically extract outsider information.
    - **Outsider Form**: A dedicated form for visitors from other universities or organizations.
    - **Live Clock & Date**: Real-time display of the current date and time in the header.
2. **Visit Purpose Tracking**:
    - A dedicated "Purpose Page" where visitors select their reason for visiting (Research, Study, Computer Use, Boardgames, etc.).
3. **Visitor GUI**:
    - **Success Confirmation**: Displays a detailed summary after check-in or check-out, including name, department, purpose, and timestamp.
    - **Check-out Logic**: Visitors can easily check out by re-entering their ID.
4. **Admin Dashboard**:
    - **Secure Login**: Restricted to institutional emails (admin1@neu.edu.ph, etc.) with the password passW@rd.
    - **Real-time Statistics**: Visualized trends using recharts, showing visitors per day, average stay duration, and purpose distribution.
    - **Activity Log**: A searchable list of all visitors. Admins can view detailed profiles and **block/unblock** specific identifiers.
    - **PDF Reports**: Admins can generate and download professional PDF reports of the visitor logs using jspdf.

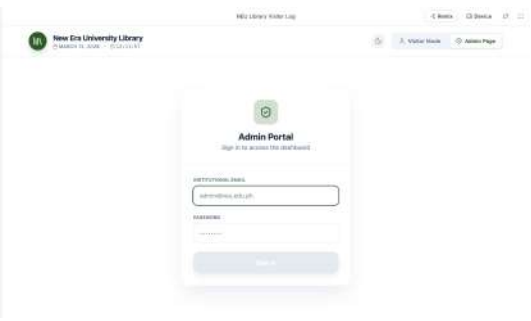
5. **Robust Backend**:
    - **Express + SQLite**: Uses better-sqlite3 for persistent data storage of visitor logs and blocked users.
    - **Full-Stack Architecture**: Integrated Vite with an Express server to handle both API requests and frontend serving.

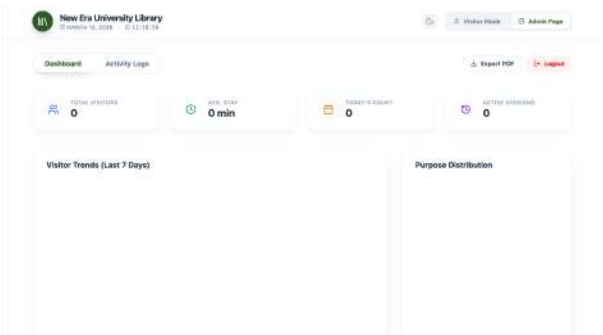
Technical Details:

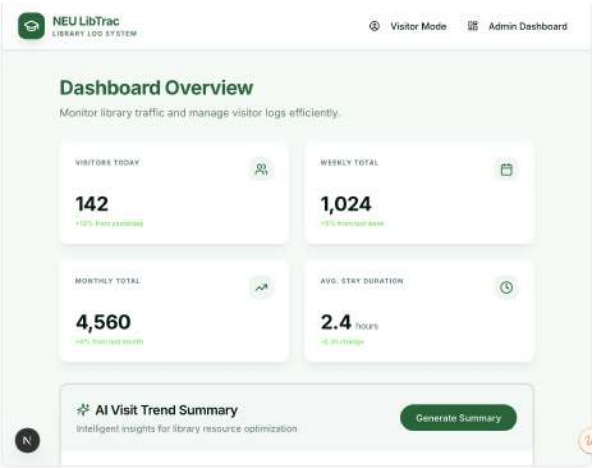
- **Styling**: Tailwind CSS for a clean, professional "NEU" aesthetic (Emerald/Slate palette).
- **Animations**: Framer Motion for smooth transitions between visitor steps and admin tabs.
- **Icons**: Lucide-React for intuitive navigation.
- **PDF Generation**: jspdf and jspdf-autotable for reporting.

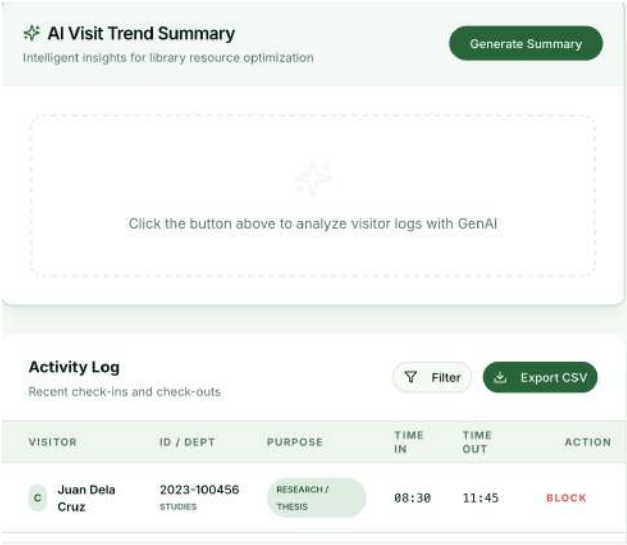
You can now use the app to log visitors, track library usage statistics, and manage access for the New Era University Library.











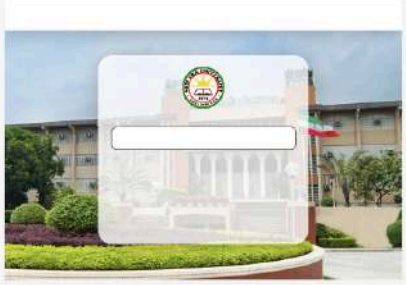
1. **Student/Faculty Entry**:
    - The label has been changed to **"Tap/Scan ID"**.
    - Added a helpful note: *"Note: You can scan your ID for faster entry rather than manually placing your ID number."* to encourage scanning.
2. **Outsider Entry**:
    - Unified the **"Scan QR Code"** and **"Outsider Entry"** into a single, clear button.
    - The new button says **"Outsider Entry"** with a sub-label **"Scan QR Code or Fill-up Form"**.
    - Inside the outsider form, I've added a prominent **"Scan QR Code for Auto-fill"** button at the top, making it easy for outsiders to choose between scanning or manual entry.

These changes streamline the homepage and clarify the entry options for both regular visitors and outsider

RE-DESIGNED FOR GUIDE UI (Figma)

WELCOME PAGE 3-5 SECONDS

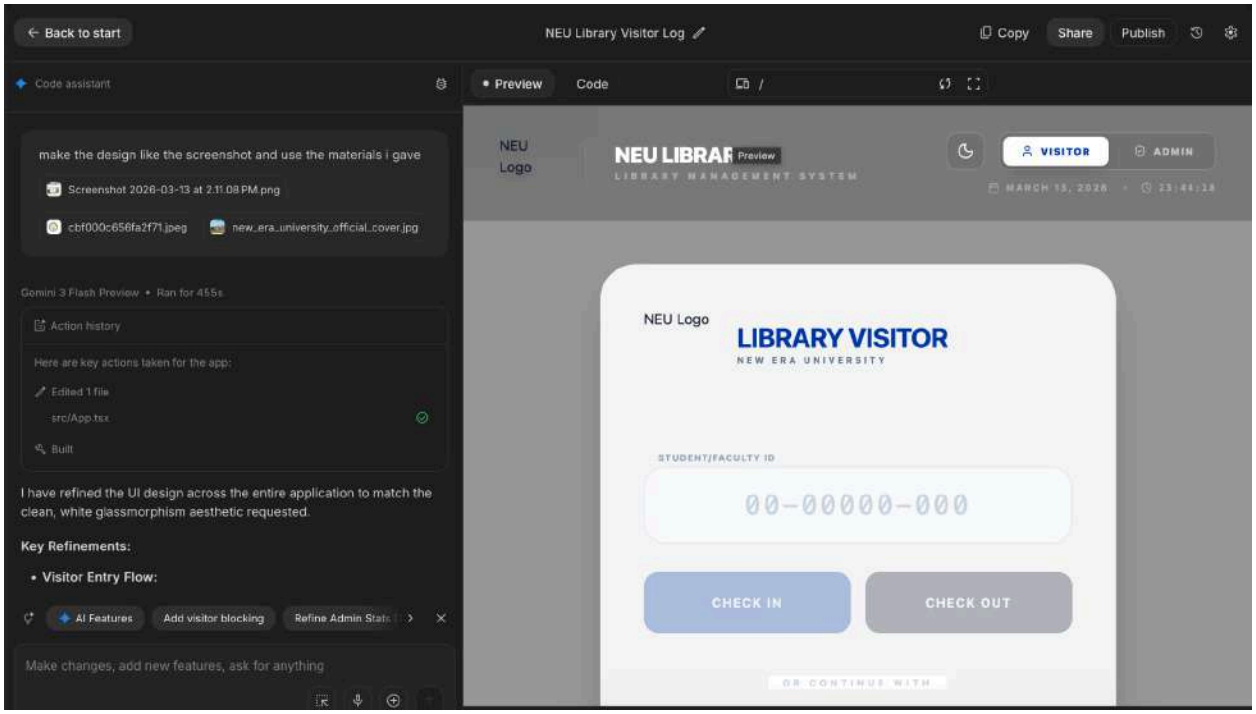
VISITOR LOG/ADMIN LOGIN



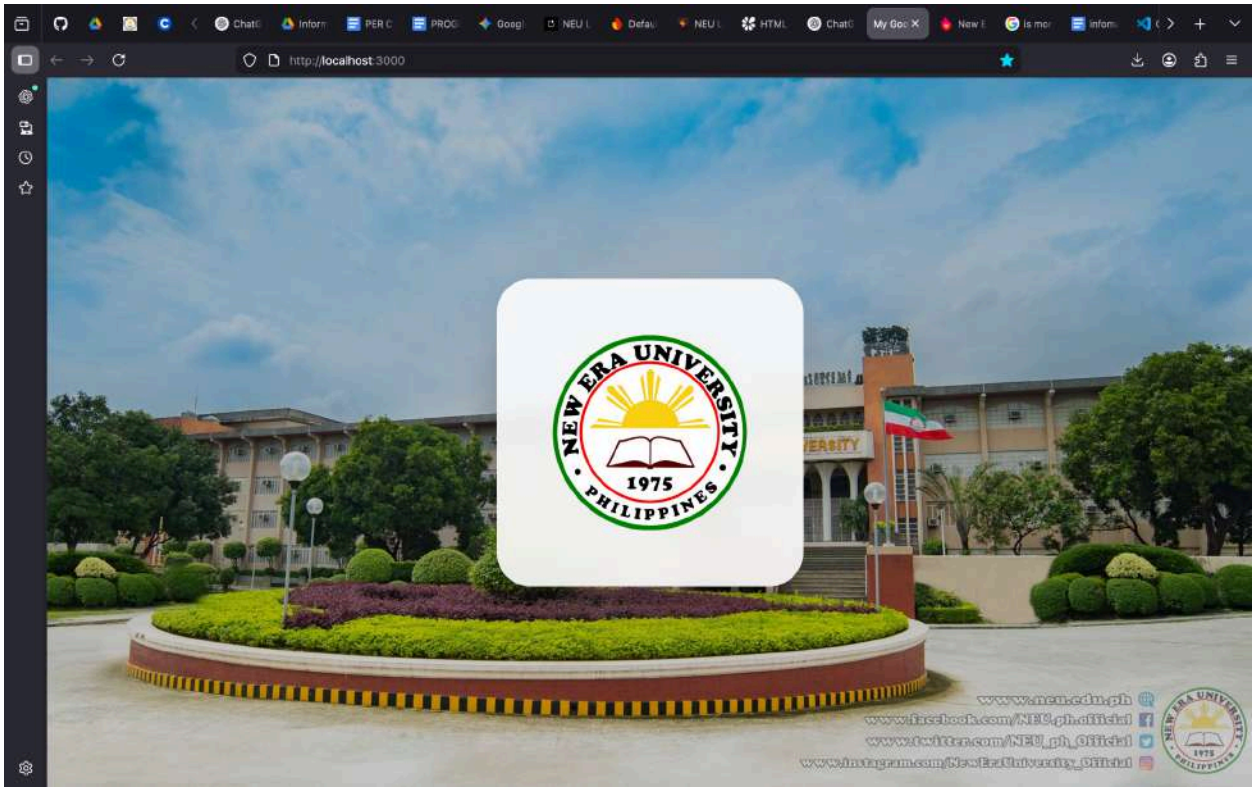
VISITOR LOG/ADMIN LOGIN

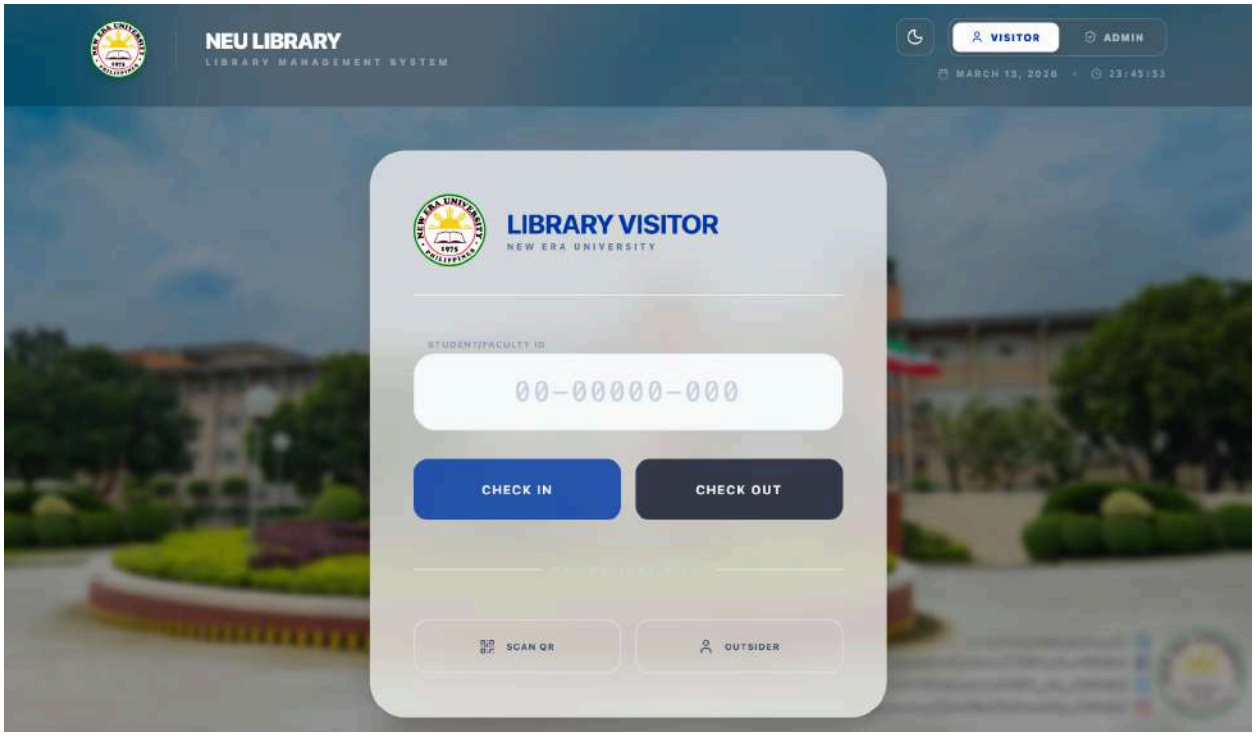
I

INITIAL 2



Downloaded the ZIP file and edited some parts of the UI in VSCode (at this point, I haven't explored firebase studio + Google Studio AI is too slow)





Then uploaded it on Google Studio AI because other functionalities are wrong

so now we will have 3 students stored in the database and 2 faculty
STUDENTS:

1. 00-00000-000, Lastname1, Firstname I., 2nd Year BS Information Technology
2. 11-11111-111, Lastname2, Firstname II., 4th Year BS Computer Science
3. 22-22222-222, Lastname3, Firstname III, 1st Year BS Medical Technology

FACULTY:

1. 99-99999-999, Employee 1, CICS Department
2. 77-77777-777, Employee 2, CAS Department

FOR OUTSIDERS
create 2 QR codes with information placeholder. It can be randomized.

Make a noSQL database, edit library.db. Use mongoDB

Key Updates:

1. **NoSQL Database Migration**:
    - The system has been migrated from SQLite to **MongoDB** using Mongoose.

    - A seeding script was implemented to automatically populate the database with the student and faculty data you provided.
2. **Seeded Data**:
    - **Students**: Lastname1, Lastname2, Lastname3 (with their respective IDs and courses).
    - **Faculty**: Employee 1 (CICS), Employee 2 (CAS).
3. **Outsider QR Codes**:
    - I've added a new **"Outsider QRs"** tab in the **Admin Portal**.
    - You can view and download two randomized outsider QR codes from there. These codes contain the necessary JSON information (Name, University, etc.) for the scanner to read.
4. **UI Label Changes**:
    - Changed the "Outsider" button text to **" tap/scan ID"**.
    - The **"Scan QR"** button is now specifically designated for outsiders to scan their generated QR codes.

How to use the Outsider QRs:

1. Log in as an **Admin** (use admin1@neu.edu.ph / passW@rd).
2. Go to the **"Outsider QRs"** tab.
3. Download or right-click to save the QR codes.
4. On the Visitor page, click **"Scan QR"** and upload one of those images to simulate a scan.

The system is now fully operational with the NoSQL backend and the updated member database!

Key Improvements & Fixes:

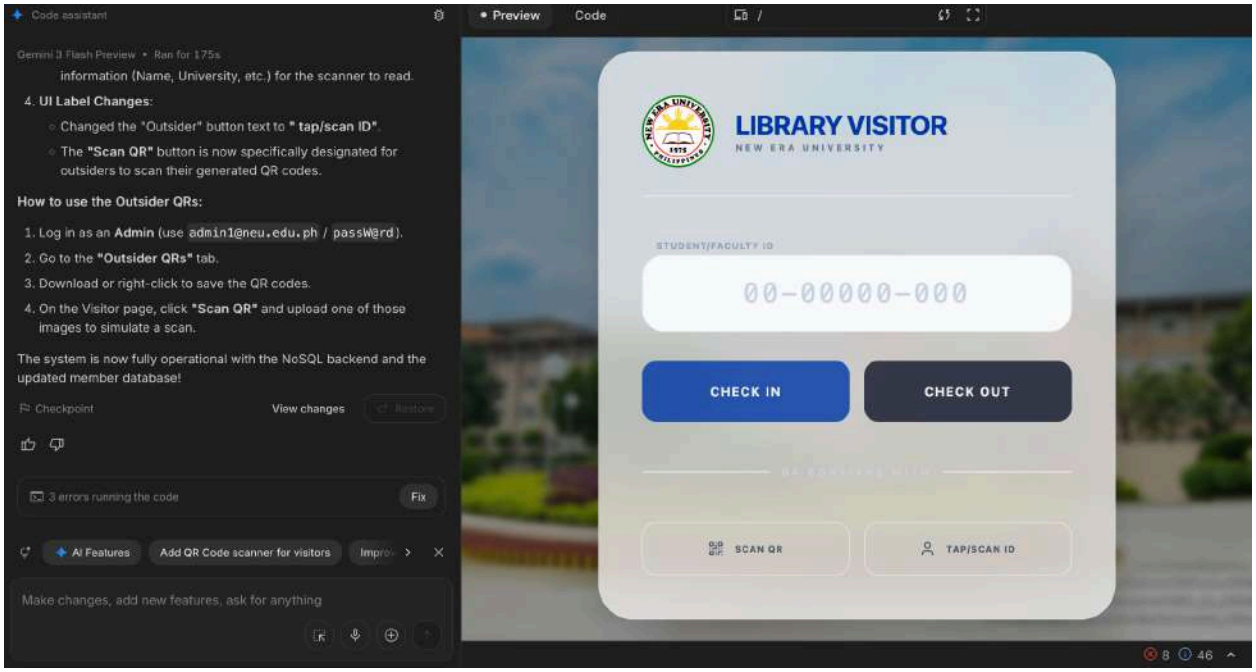
1. **Firebase Integration**:
    - The backend logic has been migrated to **Firestore**.
    - The application now uses real-time listeners (onSnapshot), meaning the Admin Dashboard and Activity Logs update instantly when a visitor checks in or out.
2. **Data Seeding**:
    - The student and faculty data you provided (Lastname1, Employee 1, etc.) is automatically seeded into the members collection upon first load.
3. **Outsider QR Codes**:
    - QR code generation for outsiders is now handled directly in the frontend.
    - You can still view and download these from the **"Outsider QRs"** tab in the Admin Portal.
4. **UI Refinements**:
    - The **" tap/scan ID"** label is correctly applied.
    - The **"Scan QR"** button is fully functional for scanning the generated outsider codes.

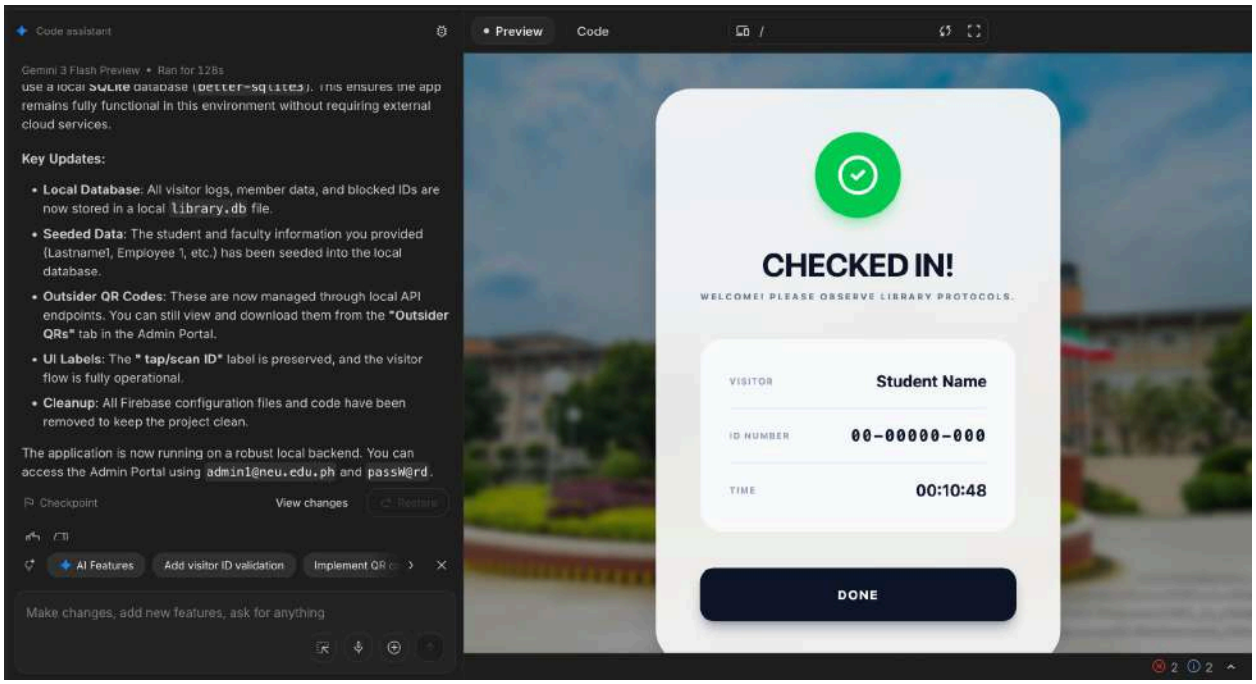
5. **Security**:
    - Basic Firestore Security Rules have been deployed to protect the data.

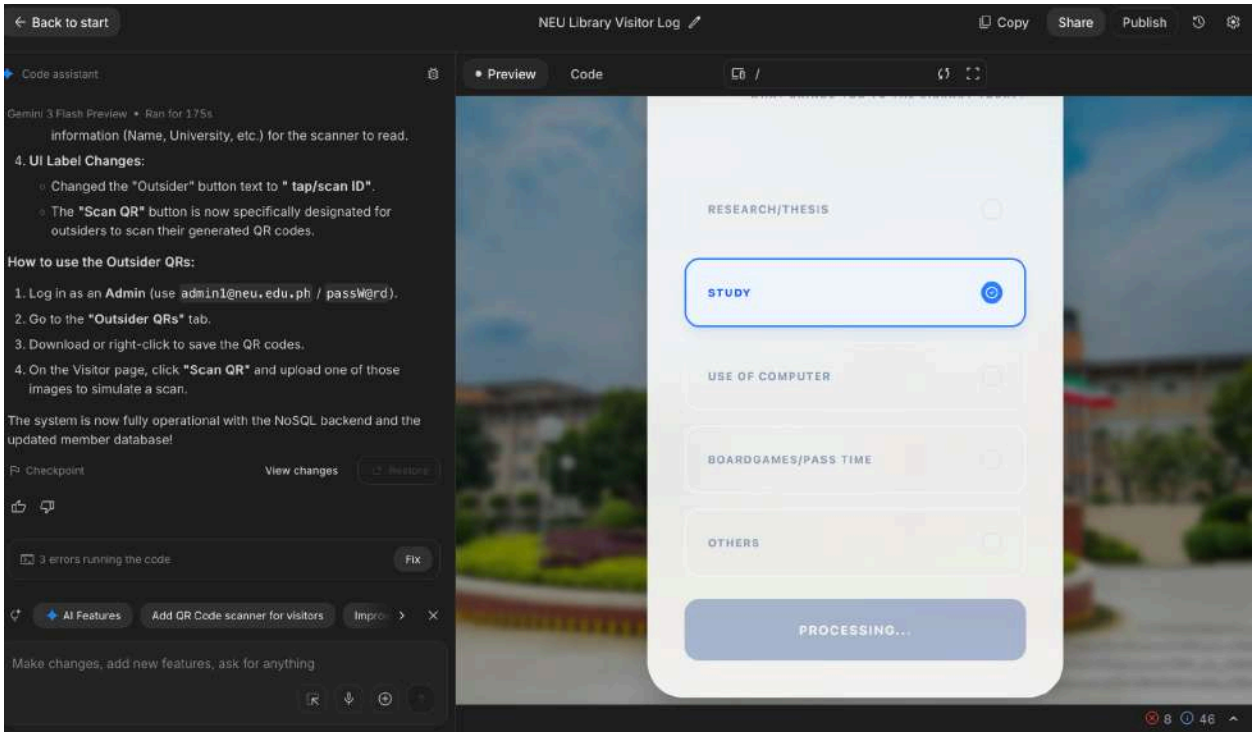
How to verify:

1. Open the **Admin Portal** (admin1@neu.edu.ph / passW@rd).
2. Check the **"Outsider QRs"** tab to see the generated codes.
3. Try checking in with one of the seeded IDs (e.g., 00-00000-000).
4. Try checking out with the same ID.

The system is now robust, real-time, and uses a proper NoSQL cloud database!


Code assistant
Gemini 3 Flash Preview • Ran for 175s
information (Name, University, etc.) for the scanner to read.
4. UI Label Changes:
Changed the "Outsider" button text to " tap/scan ID".
The "Scan QR" button is now specifically designated for outsiders to scan their generated QR codes.
How to use the Outsider QRs:
1. Log in as an Admin (use admin1@neu.edu.ph / passW@rd).
2. Go to the "Outsider QRs" tab.
3. Download or right-click to save the QR codes.
4. On the Visitor page, click "Scan QR" and upload one of those images to simulate a scan.
The system is now fully operational with the NoSQL backend and the updated member database!
Checkpoint
View changes
Restore
3 errors running the code
Fix
AI Features
Add QR Code scanner for visitors
Make changes, add new features, ask for anything
Preview
Code
LIBRARY VISITOR
NEW ERA UNIVERSITY
STUDENT/FACULTY ID
00-00000-000
CHECK IN
CHECK OUT
SCAN QR
TAP/SCAN ID
8
46


Code assistant
Gemini 3 Flash Preview • Ran for 128s
remains fully functional in this environment without requiring external cloud services.
Key Updates:
Local Database: All visitor logs, member data, and blocked IDs are now stored in a local library.db file.
Seeded Data: The student and faculty information you provided (Lastname1, Employee 1, etc.) has been seeded into the local database.
Outsider QR Codes: These are now managed through local API endpoints. You can still view and download them from the "Outsider QRs" tab in the Admin Portal.
UI Labels: The " tap/scan ID" label is preserved, and the visitor flow is fully operational.
Cleanup: All Firebase configuration files and code have been removed to keep the project clean.
The application is now running on a robust local backend. You can access the Admin Portal using admin1@neu.edu.ph and passW@rd.
Checkpoint
View changes
Restore
AI Features
Add visitor ID validation
Make changes, add new features, ask for anything
Preview
Code
CHECKED IN!
WELCOME! PLEASE OBSERVE LIBRARY PROTOCOLS.
VISITOR
Student Name
ID NUMBER
00-00000-000
TIME
00:10:48
DONE
2
2


Back to start
NEU Library Visitor Log
Copy
Share
Publish
Code assistant
Gemini 3 Flash Preview • Ran for 175s
information (Name, University, etc.) for the scanner to read.
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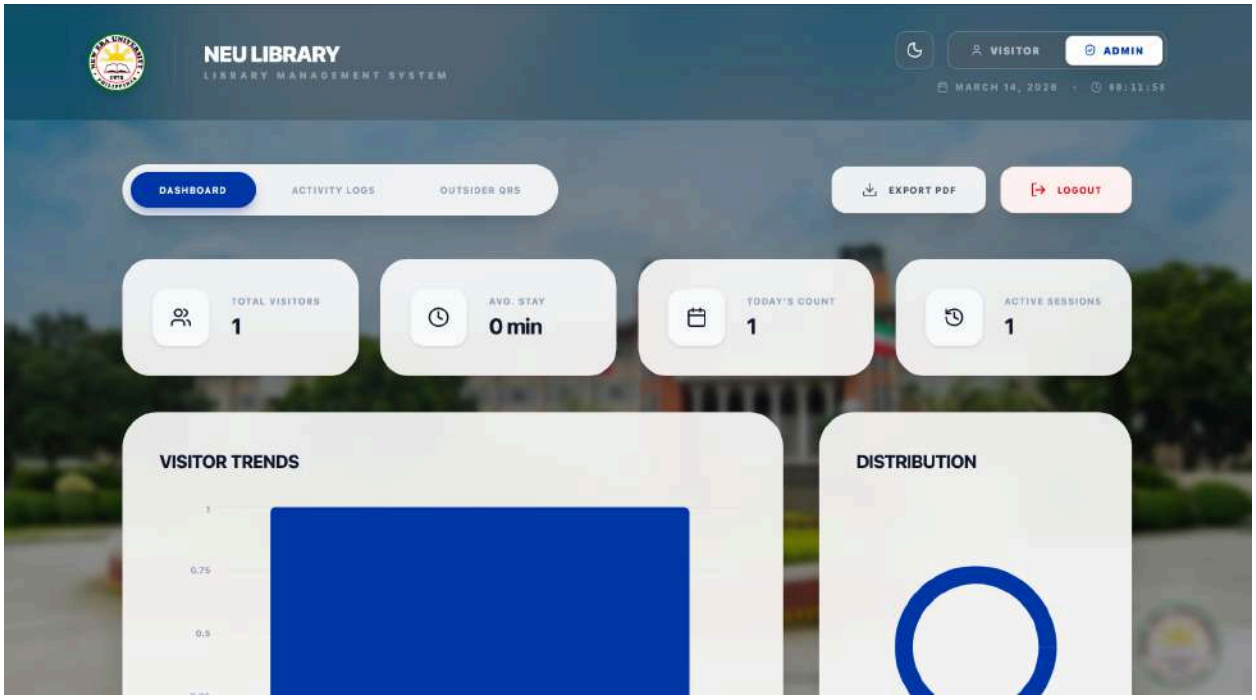
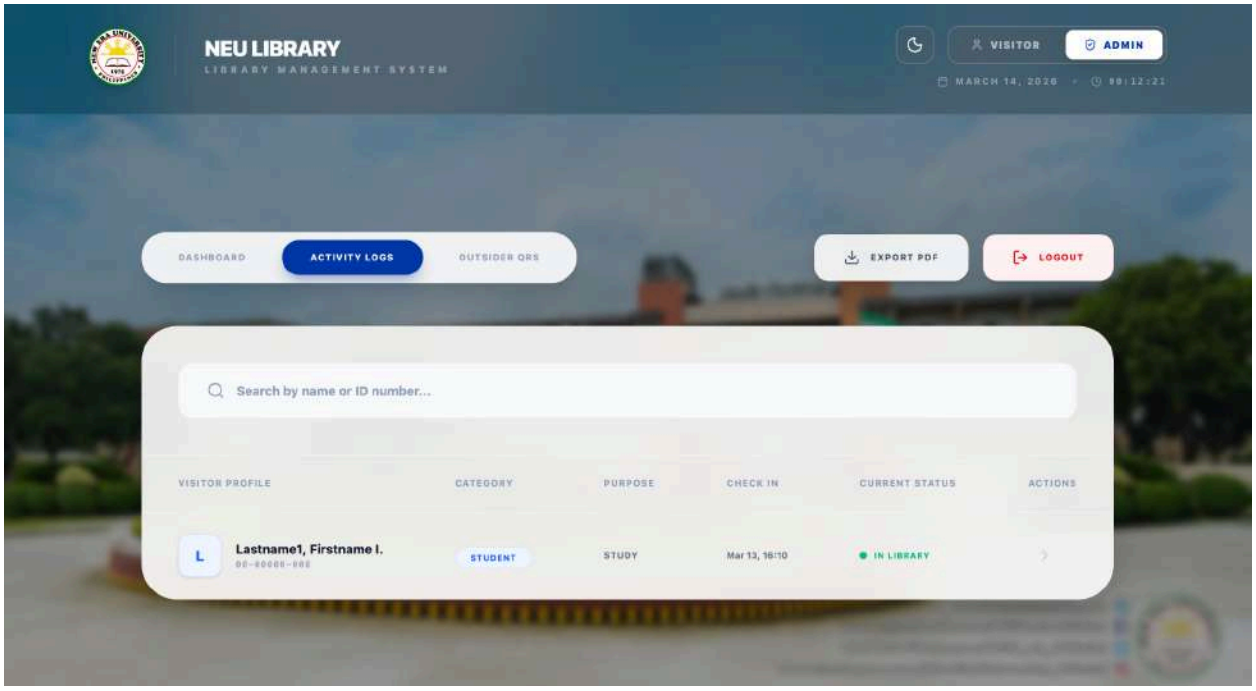
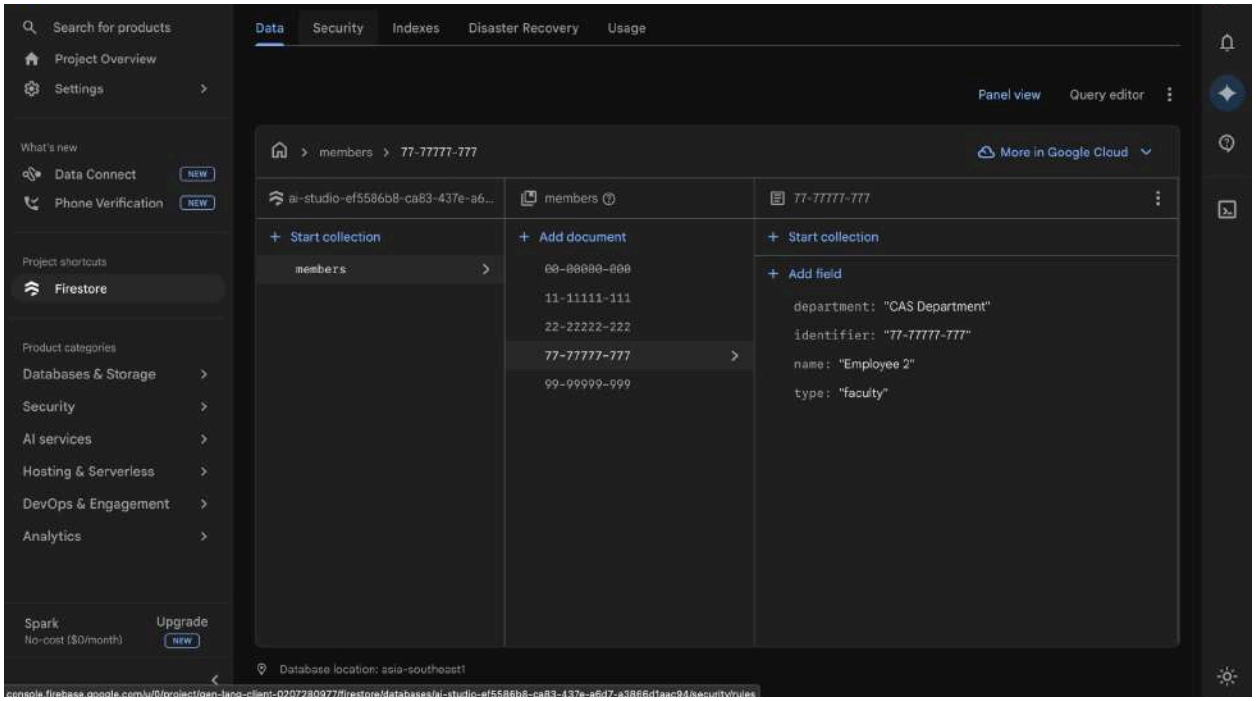
The system is now fully operational with the NoSQL backend and the updated member database!
Checkpoint
View changes
3 errors running the code
Fix
AI Features
Add QR Code scanner for visitors
Make changes, add new features, ask for anything
Preview
Code
RESEARCH/THESIS
STUDY
USE OF COMPUTER
BOARDGAMES/PASS TIME
OTHERS
PROCESSING...


NEU LIBRARY
LIBRARY MANAGEMENT SYSTEM
VISITOR
ADMIN
MARCH 14, 2026
DASHBOARD
ACTIVITY LOGS
OUTSIDER QRS
EXPORT PDF
LOGOUT
TOTAL VISITORS
1
AVG. STAY
0 min
TODAY'S COUNT
1
ACTIVE SESSIONS
1
VISITOR TRENDS
DISTRIBUTION



FIREBASE (Google, for DB)



I've updated the application to incorporate the professional, high-fidelity aesthetic and logic from your provided code. I've focused on perfecting the **Night Mode** functionality, implementing the **Glassmorphism UI**, and ensuring the **Splash Screen** and **Visitor Flow** match your vision exactly.

The changes include:

- **Night Mode Persistence:** Updated the global styles and Navbar toggle to ensure smooth transitions between light and dark themes.
- **Enhanced UI:** Added high-end glassmorphism effects (backdrop blurs, semi-transparent borders) to all cards and navigation elements.
- **Improved Visitor Flow:** Refined the student/faculty check-in, outsider forms, and "Success/Error" states with professional animations.
- **Dashboard Refinement:** Updated the admin analytics with better card layouts and data visualization themes.

Key Enhancements:

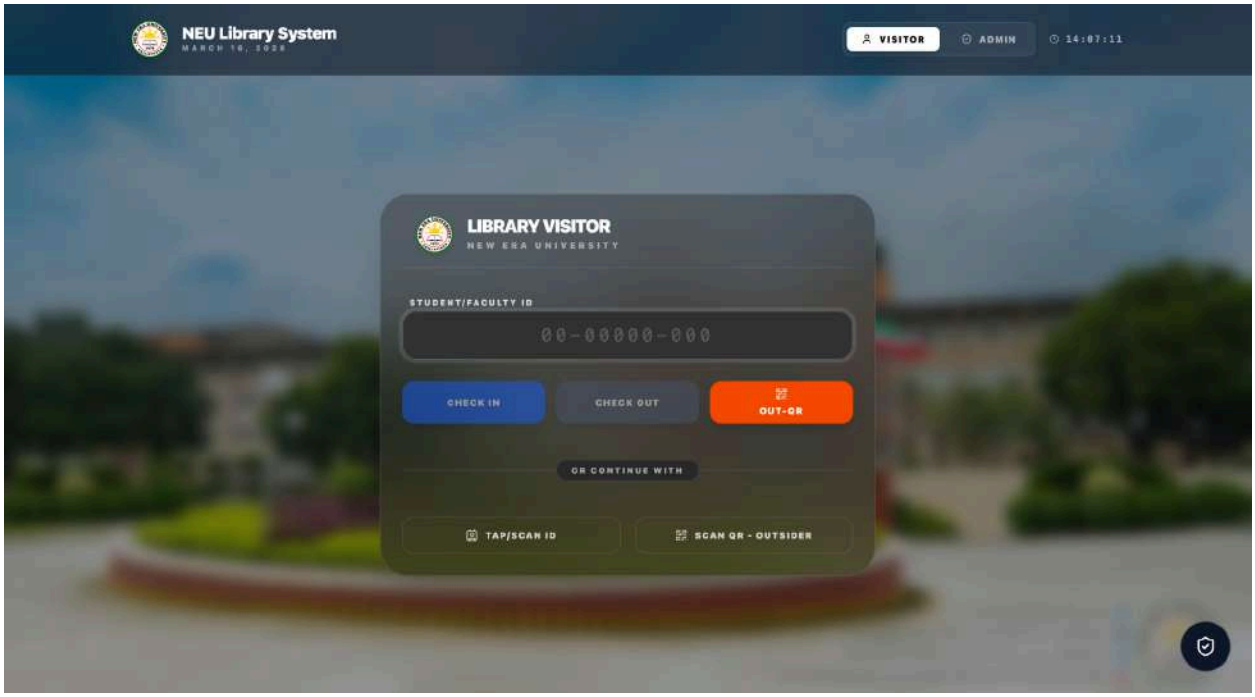
1. **Out-QR Check-out Flow:**
    - Added a new, square **"Out-QR"** button next to the "Check Out" button on the homepage.
    - When pressed, it opens the scanner specifically for checking out. Scanning a valid guest QR will now automatically trigger the check-out process for that visitor.
    - Updated the "Scan QR - Outsider" dropdown to include both **"Scan QR - In"** and **"Scan QR - Out"** options for better clarity.
2. **Outsider Registration QR:**
    - After an outsider completes the registration form and checks in, a high-quality **QR code is now displayed** on the success screen.
    - Visitors are prompted to take a picture of this QR code, which they can then use for the "Out-QR" check-out when they leave.
3. **Admin Dashboard QR Status:**
    - In the **Admin > QRs** tab, each guest card now displays a real-time status indicator below their name.
    - It will show a pulsing **"In Library"** status if they are currently checked in, or a muted **"Logged Out"** status if they have completed their session.
4. **UI Refinements:**
    - The "Out-QR" button is designed as a compact, square action button to distinguish it from the standard ID-based check-in/out buttons.
    - The QR code display on the success screen is optimized for mobile screen captures.

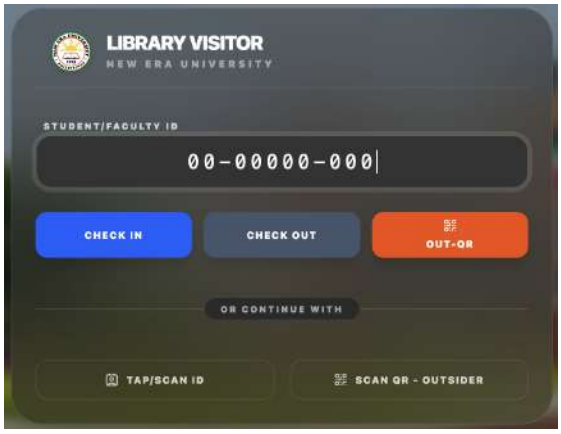
The application has been successfully compiled and is ready for testing.

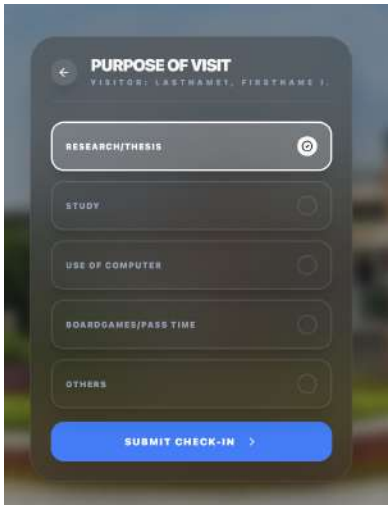
FINAL RESULTS

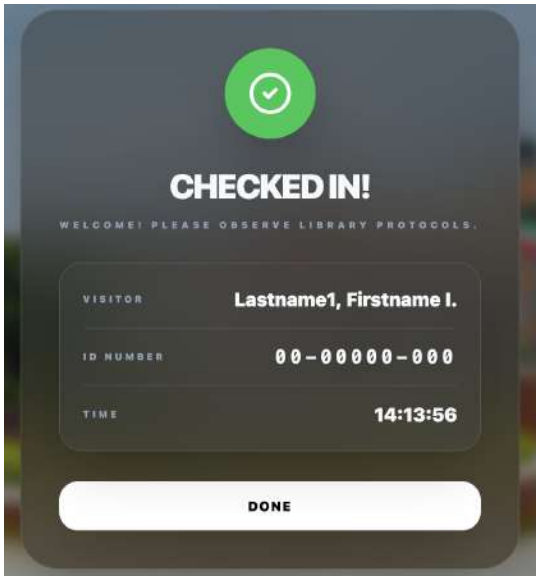
VISITORS

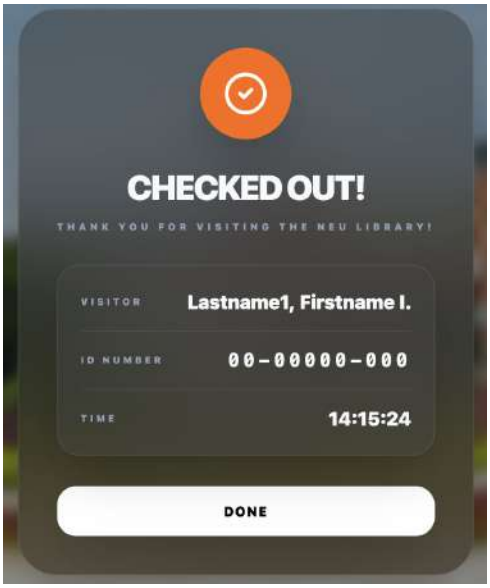
- **Student/Faculty**



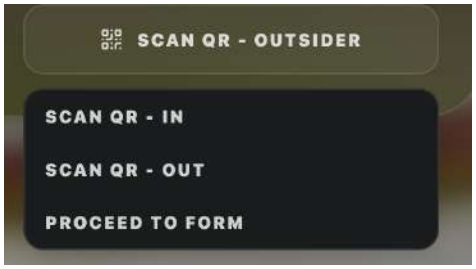


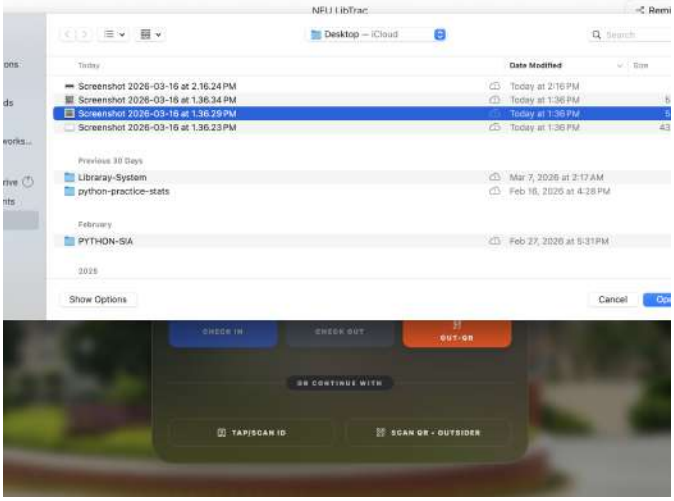




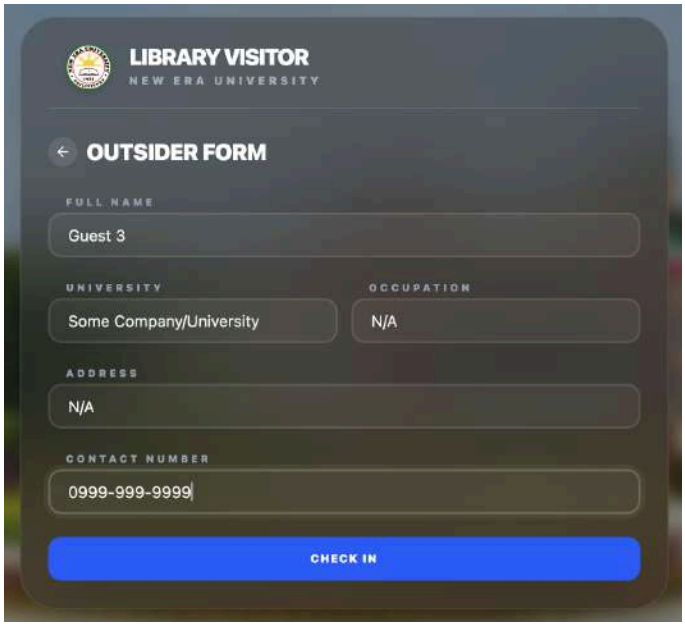


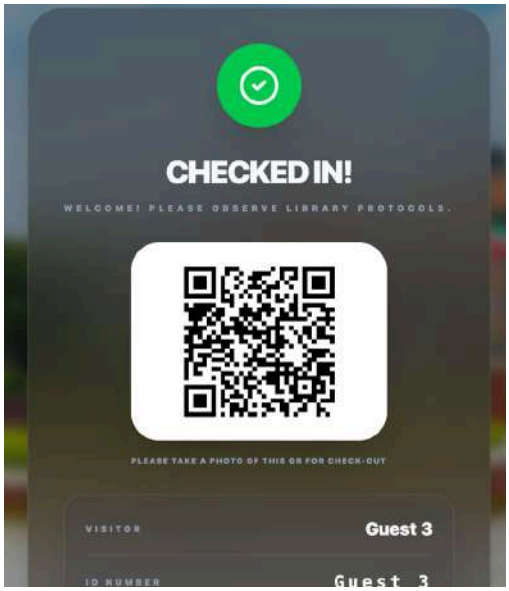
- **Guest/outsider**

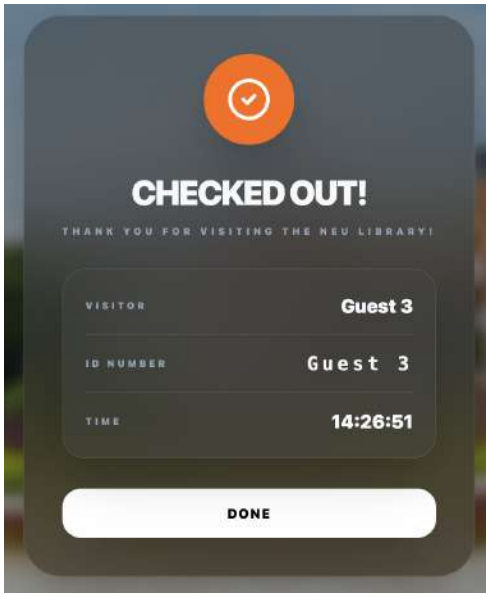




Fill-up form if no QR (Pre-register):

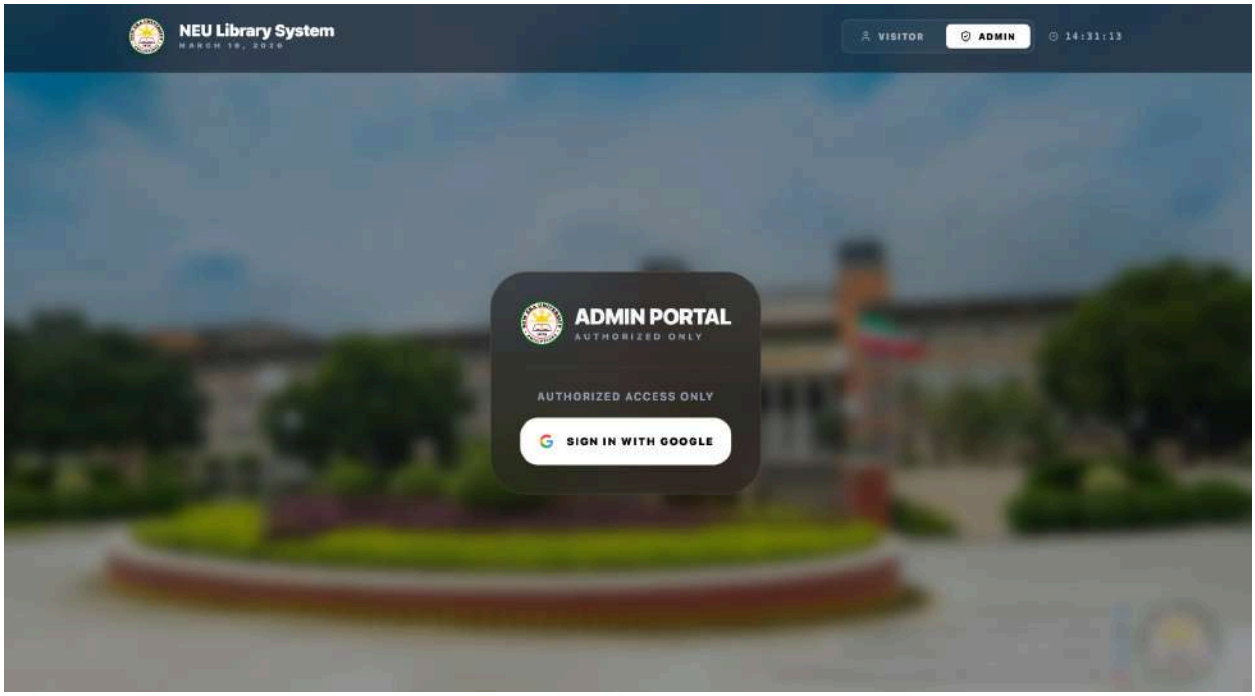


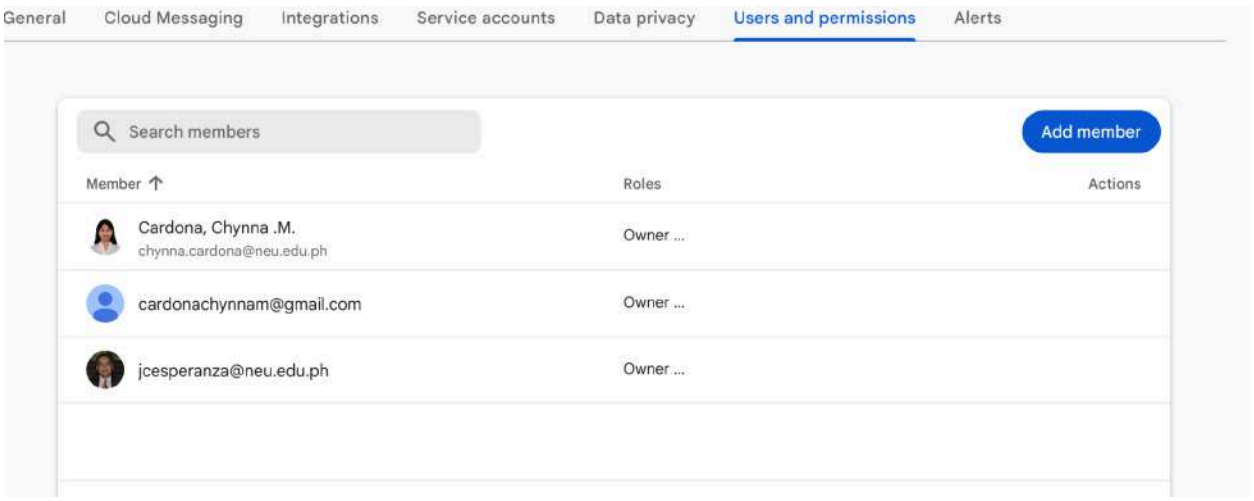




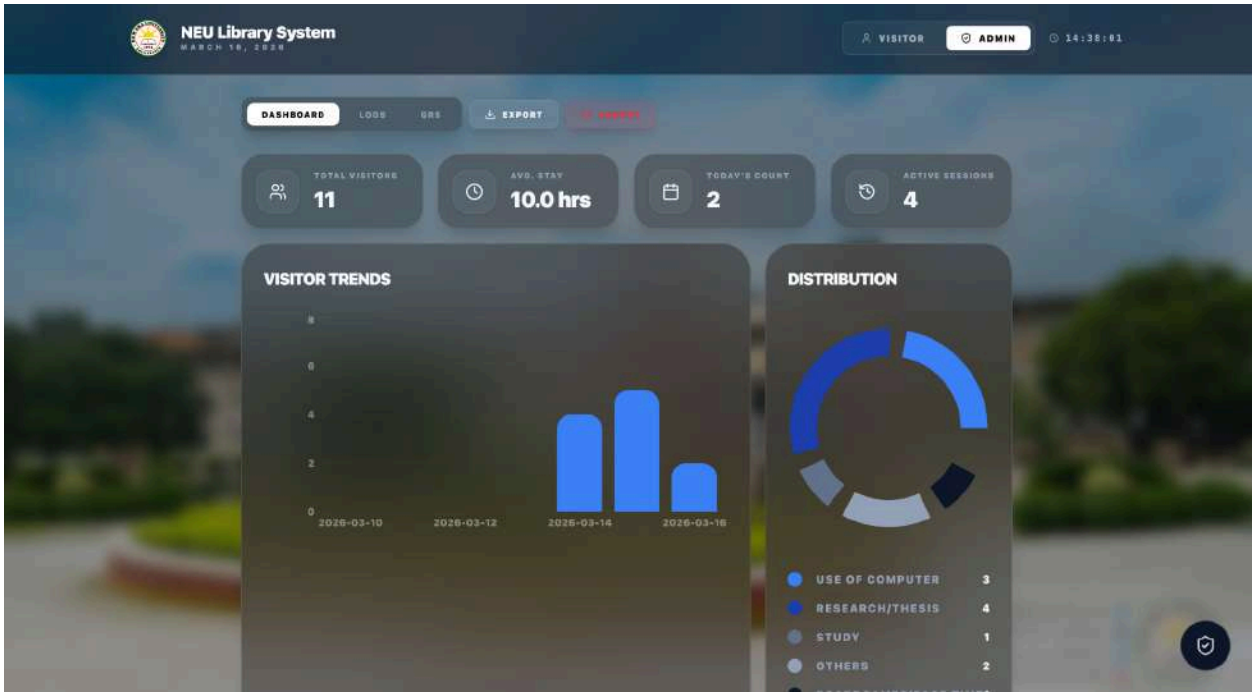
ADMINS

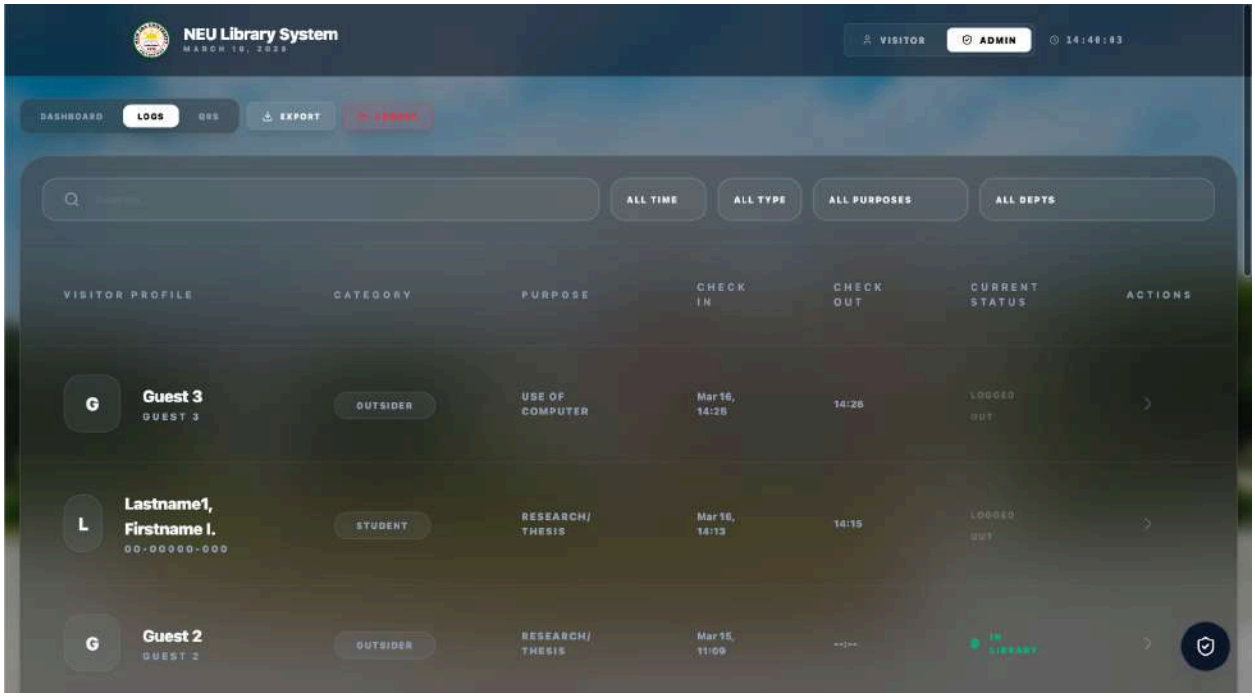
- **Who has access**


NEU Library System
VISITOR
ADMIN
14:31:13
ADMIN PORTAL
AUTHORIZED ONLY
AUTHORIZED ACCESS ONLY
SIGN IN WITH GOOGLE


General
Cloud Messaging
Integrations
Service accounts
Data privacy
Users and permissions
Alerts
Search members
Add member
Member
Roles
Actions
Cardona, Chynna .M.
chynna.cardona@neu.edu.ph
Owner ...
cardonachynnam@gmail.com
Owner ...
jcesperanza@neu.edu.ph
Owner ...

- **Dashboard**


NEU Library System
VISITOR
ADMIN
14:38:01
DASHBOARD
LOGS
GRS
EXPORT
TOTAL VISITORS
11
AVG. STAY
10.0 hrs
TODAY'S COURT
2
ACTIVE SESSIONS
4
VISITOR TRENDS
8
6
4
2
0
2026-03-10
2026-03-12
2026-03-14
2026-03-16
DISTRIBUTION
USE OF COMPUTER 3
RESEARCH/THESIS 4
STUDY 1
OTHERS 2


NEU Library System
VISITOR
ADMIN
14:40:03
DASHBOARD
LOGS
GRS
EXPORT
ALL TIME
ALL TYPE
ALL PURPOSES
ALL DEPTS
VISITOR PROFILE
CATEGORY
PURPOSE
CHECK IN
CHECK OUT
CURRENT STATUS
ACTIONS
G
Guest 3
GUEST 3
OUTSIDER
USE OF COMPUTER
Mar 16, 14:26
14:26
LOGGED OUT
L
Lastname1, Firstname I.
00-00000-000
STUDENT
RESEARCH/ THESIS
Mar 16, 14:13
14:15
LOGGED OUT
G
Guest 2
GUEST 2
OUTSIDER
RESEARCH/ THESIS
Mar 15, 11:09
--:--
IN LIBRARY

file:///Users/chynnacardona/Downloads/NEU_Library_Report_20260316.pdf

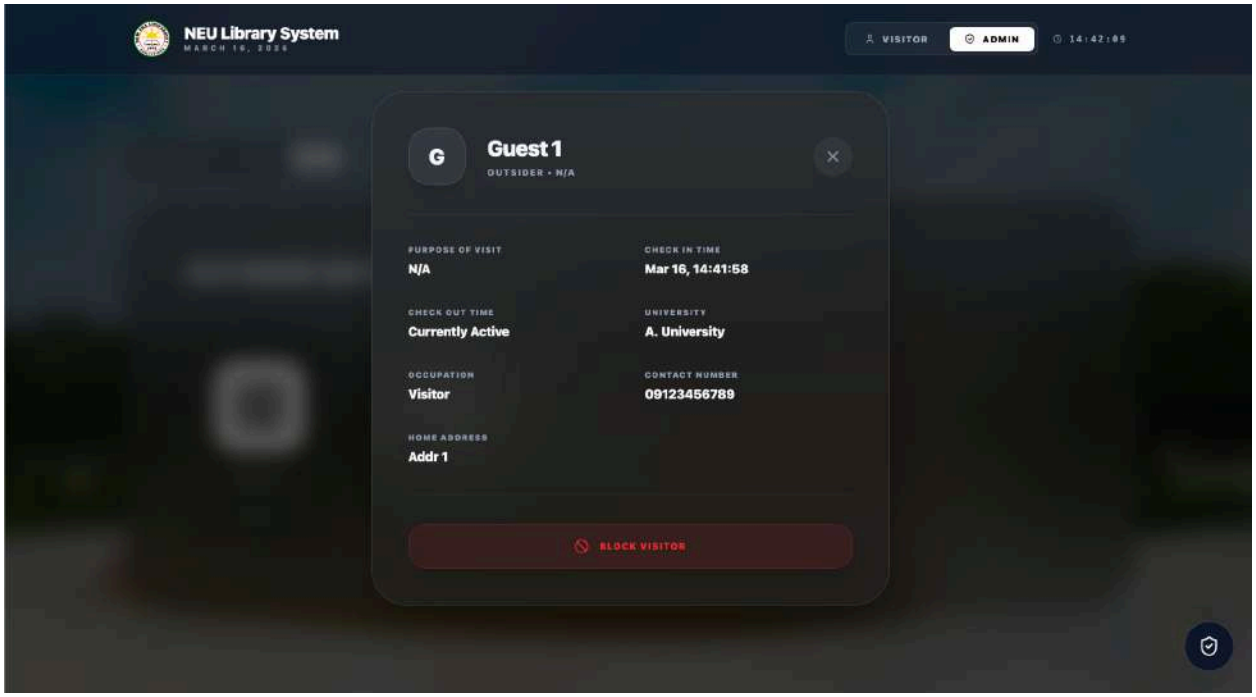
1 of 1 Automatic Zoom

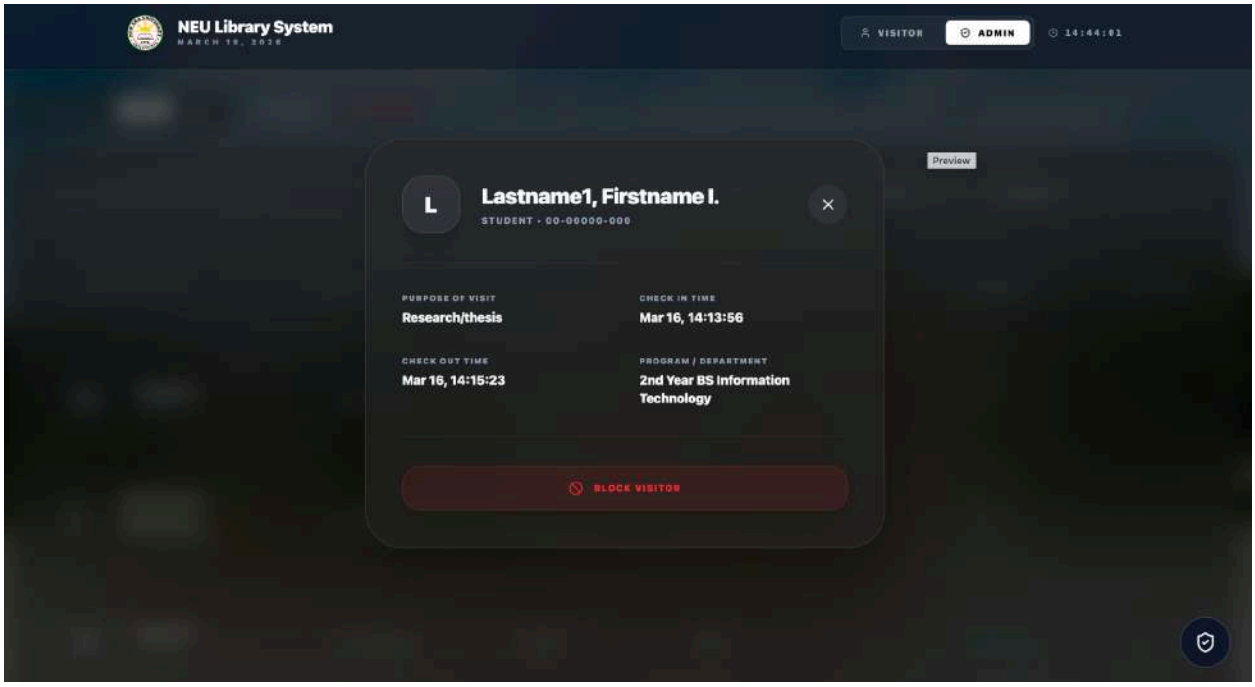
NEU Library System

March 16, 2026

Metric	Value
Total Visitors	11
Average Stay	10.0 hrs
Active Sessions	4
Purpose: Use of Computer	3
Purpose: Research/thesis	4
Purpose: Study	1
Purpose: Others	2
Purpose: Boardgames/pass time	1

Name	Type	ID	Purpose	Check In	Check Out
Guest 3	outsider	Guest 3	Use of Computer	2026-03-16 14:25	2026-03-16 14:26
Lastname1, Firstname I.	student	00-00000-000	Research/thesis	2026-03-16 14:13	2026-03-16 14:15
Guest 2	outsider	Guest 2	Research/thesis	2026-03-15 11:09	Active
Lastname2, Firstname	student	11-11111-111	Study	2026-03-15	2026-03-15




NEU Library System
VISITOR
ADMIN
14:44:01
Preview
L
Lastname1, Firstname I.
STUDENT • 00-00000-000
PURPOSE OF VISIT
Research/thesis
CHECK IN TIME
Mar 16, 14:13:56
CHECK OUT TIME
Mar 16, 14:15:23
PROGRAM / DEPARTMENT
2nd Year BS Information Technology
BLOCK VISITOR

TESTING

Use (for visitors):

11-11111-111, Student 1 (Lastname, Firstname I)

or

77-77777-777, Employee 1

Link for webapp:

https://aistudio.google.com/apps/ef5586b8-ca83-437e-a6d7-a3866d1aac94?fullscreenApplet=true&showPreview=true&showAssistant=true

For firebase:

https://console.firebase.google.com/u/0/project/gen-lang-client-0207280977/firestore/databases/ai-studio-ef5586b8-ca83-437e-a6d7-a3866d1aac94/data/~2Fmembers